ANIMAL WELFARE & ECOSYSTEM RESTORATION

Tierra del Fuego

A restoration decision-support platform for the Culebra wildfire

Group N ANWE



Product image supplied by Group N ANWE

DEMO CASE

Sierra de la Culebra, Spain

EVIDENCE SOURCES

CEMS, Sentinel-2 and GBIF

DECISION STATUS

Unreviewed draft / field checks required

01. Relevance

Restoration decisions after wildfire

June 2022 Culebra case
Animal needs remain a field question

The decision challenge

Limited animal evidence

Historical records cannot establish current animal use or welfare.

Competing habitat needs

Planting and machinery may disturb cover, open habitat or breeding sites.

Uncertain recovery

Satellite change alone does not establish habitat recovery speed.

Tierra del Fuego approach

Traceable environmental evidence

Combine CEMS mapping, Sentinel imagery and dated GBIF records.

Explicit animal concerns

Inspect food, water, shelter, movement and breeding for six groups.

Conditional options and review

Compare restoration pathways and record what needs field verification.

02. Approach

Evidence and decision workflow

1. **Retrieve real evidence**

   CEMS footprint observed 25 June 2022. Satellite pair: 10 June / 5 July 2022.

2. **Compare matched windows**

   GBIF observations: June 2021 / June 2023, all groups, 5 km outside buffer.

3. **Interpret and document**

   AI draft, five values, conditional options and a PDF.

Uncertain locations: 143 / 204 records. Buffer is not a validated unaffected control.

Actual demo results

Greenness (NDVI): 0.482 / 0.385
Spectral change (dNBR): 0.114
Inside footprint: 13 / 0 records
Nearby buffer: 35 / 5 records

Historical records / no population inference

03. Ethical Architecture

Five values and multispecies concerns

⚖️ Design for Values

Animal welfare and ecological recovery

Protect animal needs while investigating habitat functions and regeneration.

Transparency and uncertainty

Show sources, methods, sampling bias and missing field measurements.

Accountability

Humans justify decisions. A model draft cannot approve restoration.

Questions that need evidence

Disturbance and conflicting habitats

Which animals use open habitat, cover and refuges? Check breeding timing.

Privacy and reporting bias

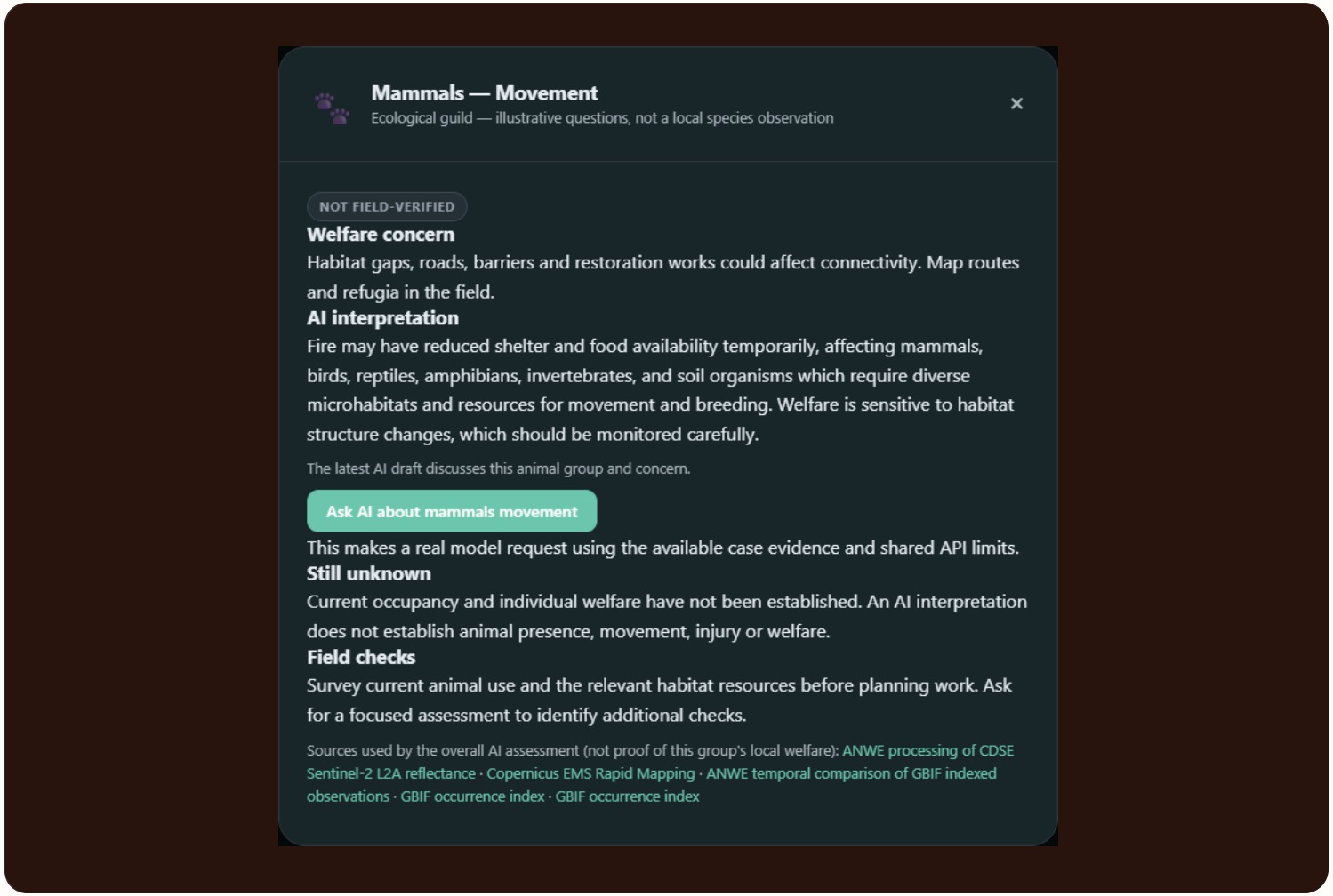
Withhold exact animal coordinates. Missing observations do not prove absence.

Evidence gaps across animal groups

Consider mammals, birds, reptiles, amphibians, invertebrates and soil biota.

04. Interface

Animal welfare and AI in the demo



Actual mammals movement popup / real session draft / no new model call for these slides

Actual model response

gpt-4.1-mini considers the five values and three restoration options.

Field questions remain visible

Each concern shows hypotheses, unknown conditions and checks. Current occupancy remains unverified.

A report for restoration decisions

RESTORATION TEAMS

Compare conditional options

Natural regeneration, assisted restoration and active restoration. Survey animal use and disturbance.

LAND MANAGERS

Document the decision

28-page report with actual evidence, AI draft, full retrieved species lists, methods and glossary.

ECOLOGISTS & FIELD STAFF

Verify conditions before action

Check refuges, connectivity, breeding, regeneration and erosion. Qualified review and permissions follow.